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Professional Experience

2024 – Present Associate Professor, Rutgers University
2018 – 2024 Assistant Professor, Rutgers University
2015 – 2018 Assistant Research Professor, Rutgers University
2017 Fulbright Scholar, University of Göthenburg Sweden
2014 – 2015 Postdoctoral Research Associate, Rutgers University
2009 – 2014 Graduate Research Assistant, Rutgers University
2007 – 2009 Graduate Research Assistant, North Carolina State University

Education

2009 – 2014 Ph.D., Rutgers University - Physical Oceanography
2007 – 2009 M.S., North Carolina State University (NCSU) - Physical Oceanography
2003 – 2007 B.S., North Carolina State University – Marine Sciences and Meteorology

External Awards & Honors

2024 The Oceanography Society Ocean Observing Team Award
2018 Marine Technology Society Young Professional Award
2017 Fulbright Scholar: University of Göthenburg Sweden

Peer-Reviewed Publications (*student)

1. *Hughes, C., Liu, Y., Lahrach, Y., Engdahl, J., Warren, H., Lee, D., Ramos, F., **Miles, T.**, Abraham, I., (2026) Asymptotically Optimal Ergodic Coverage on Generalized Motion Fields (*Preprint*) <https://arxiv.org/abs/2605.13442>
2. Deng, Y., Wu, T., Gradone, J., Wilson, W.D., **Miles, T.N.**, He, R., (2026) ENSO-Driven Modulation of the Caribbean Subsurface Salinity Maximum, *Submitted to Nature Communications Earth & Environment*, [10.31223/X5Q49Q](https://doi.org/10.31223/X5Q49Q) (preprint)
3. Pareja-Roman, LF, Engdahl J., **Miles, T.**, McGarigal C., (2026) Ocean mixing from offshore wind farms: implications for the US Mid-Atlantic Bight cold pool. *Frontiers in Marine Science*, 13, [10.3389/fmars.2026.1804138](https://doi.org/10.3389/fmars.2026.1804138)
4. Gallagher, K., Baumgartner, M., Kohut, J., **Miles, T.**, Flagg, C., McSweeney, J., et al. (2025). Passive acoustic monitoring of baleen whales using autonomous gliders in relation to offshore wind energy areas in the New York Bight. *Endangered Species Research*, 58, 257–273.
5. *Tsei, S., Howden, S., Diercks, A.-R., Zhang, J. A., **Miles, T. N.**, Nyadjro, E., et al. (2025). Low Salinity, High Ocean Heat Content, and Warm Core Eddy Effects on the Upper Ocean Response During Hurricane Sally (2020): An Analysis of a Hurricane

- Glider Observations and Coupled Atmosphere-Ocean Model. *Journal of Geophysical Research: Oceans* 130, e2024JC021470. doi: [10.1029/2024JC021470](https://doi.org/10.1029/2024JC021470)
6. *Gradone, J., **Miles, T.N.**, Palter, J.B., Glenn, S.M., Wislon, W.D., Warming and salinity changes of the upper ocean Caribbean Through-Flow since 1960, accepted for Nature Scientific Reports
 7. Beaird, N., Miller, M., Book, J., Edwards, C. R., Edwards, J., Gong, D., Lynch, S., Miles, T., Osborne, J., (2025). Subtropical Mode Water as an acoustic waveguide at the Gulf Stream. *J. Acoust. Soc. Am.* 157, A95. doi: [10.1121/10.0037521](https://doi.org/10.1121/10.0037521)
 8. *Ye, F., Miles, T., and Aziz Ezzat, A. (2025). Improved spatio-temporal offshore wind forecasting with coastal upwelling information. *Applied Energy* 380, 125010. doi: [10.1016/j.apenergy.2024.125010](https://doi.org/10.1016/j.apenergy.2024.125010)
 9. Feehan, C. J., Filbee-Dexter, K., Thomsen, M. S., Wernberg, T., and Miles, T. (2024). Ecosystem damage by increasing tropical cyclones. *Commun Earth Environ* 5, 1–10. doi: [10.1038/s43247-024-01853-2](https://doi.org/10.1038/s43247-024-01853-2)
 10. Pareja-Roman, L. F., Miles, T., and Glenn, S. (2024). Coastal upwelling modulates winds and air-sea fluxes, impacting offshore wind energy. *Front. Energy Res.* 12, 1470712. doi: [10.3389/fenrg.2024.1470712](https://doi.org/10.3389/fenrg.2024.1470712)
 11. Aristizábal Vargas, M. F., Kim, H.-S., Le Hénaff, M., **Miles, T.**, Glenn, S., & Goni, G.. Evaluation of the ocean component on different coupled hurricane forecasting models using upper-ocean metrics relevant to air-sea heat fluxes during Hurricane Dorian (2019) (2024). *Frontiers in Earth Science*, 12. <https://doi.org/10.3389/feart.2024.1342390>
 12. *Horwitz, R.A., D. Munroe, J.T., Kohut, **T. Miles**, Overlap between the Mid-Atlantic Bight Cold Pool and Offshore Wind Energy Areas, (2023) *ICES Journal of Marine Science*, <https://doi.org/10.1093/icesjms/fsad190>
 13. Ye, F., Brodie, J., **Miles, T.N.**, Ezzat, A.A., AIRU-WRF: A Physics-Guided Spatio-Temporal Wind Forecasting Model and its Application to the U.S. North Atlantic Offshore Wind Energy Areas, (2023) *Journal of Renewable Energy*, 223, 119934. <https://doi.org/10.1016/j.renene.2023.119934>
 14. **Miles, T.N.**, *Coakley, S., Engdahl, J., Rudzin, J., Tsei, S., Glenn, S. Ocean mixing during Hurricane Ida (2021): The impact of a freshwater barrier layer, (2023) Special Issue of *Frontiers in Marine Science* The Impact of Oceans on Extreme Weather Events (Tropical Cyclones), <https://doi.org/10.3389/fmars.2023.1224609>
 15. *Gradone, J.C., Wilson, W.D., Glenn, S.M., **Miles, T.N.**, 2023. Upper Ocean Transport in the Anegada Passage from Multi-Year Glider Surveys. *Journal of Geophysical Research: Oceans*. <https://doi.org/10.1029/2022jc019608>
 16. *Gentil, M., Estournel, C., Durrieu de Madron, X., Many, G., **Miles, T.**, Marsaleix, P., et al. (2022). Sediment dynamics on the outer-shelf of the Gulf of Lions during a storm: An approach based on acoustic glider and numerical modeling. *Continental Shelf Research*, 240, 104721. <https://doi.org/10.1016/j.csr.2022.104721>

17. *Wang, H., Gong, D., Friedrichs, M. A. M., Harris, C. K., **Miles, T.**, Yu, H.-C., & Zhang, Y. (2022). A Cycle of Wind-Driven Canyon Upwelling and Downwelling at Wilmington Canyon and the Evolution of Canyon-Upwelled Dense Water on the MAB Shelf. *Frontiers in Marine Science*, 9, 866075. <https://doi.org/10.3389/fmars.2022.866075>
18. Friedland, K. D., **Miles, T.**, Goode, A. G., Powell, E. N., & Brady, D. C. (2022). The Middle Atlantic Bight Cold Pool is warming and shrinking: Indices from in situ autumn seafloor temperatures. *Fisheries Oceanography*, 31(2), 217–223. <https://doi.org/10.1111/fog.12573>
19. **Miles, T.** et al. (2021), Uncrewed Ocean Gliders and Saildrones Support Hurricane Forecasting and Research, *Oceanography*, 34(4), 78–81, doi:10.5670/oceanog.2021.supplement.02-28.
20. **Miles, T.**, *S. Murphy, J. Kohut, S. Borsetti, and D. Munroe (2021), Offshore wind energy and the mid-Atlantic cold pool: A review of potential interactions, *Mar. Technol. Soc. J.*, 55(4), 72–87, doi:10.4031/MTSJ.55.4.8.
21. **Miles, T.**, W. Slade, and S. Glenn (2021), Sediment resuspension and transport from a glider integrated Laser In Situ Scattering and Transmissometry (LISST) particle analyzer, *J. Atmos. Ocean. Technol.*, 38(8), 1325–1341, doi:10.1175/JTECH-D-20-0207.1.
22. Wilson, W. D., S. Glenn, **T. Miles**, A. Knap, and C. Toro (2021), Transformative ocean observing for hurricane forecasting, readiness, and response in the caribbean tropical storm corridor, *Mar. Technol. Soc. J.*, 55(3), 90–91, doi:10.4031/MTSJ.55.3.43.
23. Le Hénaff, M., R. Domingues, G. Halliwell, J. A. Zhang, H. S. Kim, M. Aristizabal, **T. Miles**, S. Glenn, and G. Goni (2021), The Role of the Gulf of Mexico Ocean Conditions in the Intensification of Hurricane Michael (2018), *J. Geophys. Res. Ocean.*, 126(5), 1–28, doi:10.1029/2020JC016969.
24. Murphy, S. C., L. J. Nazzaro, J. Simkins, M. J. Oliver, J. Kohut, M. Crowley, and **T. N. Miles** (2021), Persistent upwelling in the Mid-Atlantic Bight detected using gap-filled, high-resolution satellite SST, *Remote Sens. Environ.*, 262, 112487, doi:10.1016/j.rse.2021.112487.
25. Optis, M., A. Kumler, J. Brodie, and **T. Miles**, (2021): Quantifying sensitivity in numerical weather prediction-modeled offshore wind speeds through an ensemble modeling approach. *Wind Energy*, <https://doi.org/10.1002/we.2611>.
26. Friedland, K. D., R. E. Morse, J. P. Manning, D. C. Melrose, **T. Miles**, A. G. Goode, D. C. Brady, J. T. Kohut, and E. N. Powell (2020), Trends and change points in surface and bottom thermal environments of the US Northeast Continental Shelf Ecosystem, *Fish. Oceanogr.*, 29(5), 396–414, doi:10.1111/fog.12485.
27. Xu, Y., **T. Miles**, and O. Schofield (2020), Physical processes controlling chlorophyll-a variability on the Mid-Atlantic Bight along northeast United States, *J. Mar. Syst.*, 103433, doi:10.1016/j.jmarsys.2020.103433.
28. Lim, H. S., **T. N. Miles**, S. Glenn, D. Kim, M. Kim, J. S. Shim, I. Chun, and K. N. Hwang (2020), Rapid Ocean Destratification by Typhoon Soulik over the Highly

- Stratified Waters of West Jeju Island, Korea, *J. Coast. Res.*, 95(sp1), 1480–1484, doi:10.2112/SI95-285.1.
29. *Wright-Fairbanks, E. K., **T. N. Miles**, W. J. Cai, B. Chen, and G. K. Saba (2020), Autonomous Observation of Seasonal Carbonate Chemistry Dynamics in the Mid-Atlantic Bight, *J. Geophys. Res. Ocean.*, 125(11), e2020JC016505, doi:10.1029/2020JC016505.
 30. Domingues, R. et al. (2019), Ocean Observations in Support of Studies and Forecasts of Tropical and Extratropical Cyclones , *Front. Mar. Sci.* , 6, 446.
 31. Testor, P. et al. (2019), OceanGliders: A Component of the Integrated GOOS , *Front. Mar. Sci.* , 6, 422.
 32. Saba, G. K., E. *Wright-Fairbanks, B. Chen, W.-J. Cai, A. H. Barnard, C. P. Jones, C. W. Branham, K. Wang, and **T. Miles** (2019), The Development and Validation of a Profiling Glider Deep ISFET-Based pH Sensor for High Resolution Observations of Coastal and Ocean Acidification , *Front. Mar. Sci.* , 6, 664.
 33. *Kobelt, J. N., W. C. Sharp, **T. N. Miles**, and C. J. Feehan (2019), Localized Impacts of Hurricane Irma on *Diadema antillarum* and Coral Reef Community Structure, Estuaries and Coasts, doi:10.1007/s12237-019-00665-4.
 34. Parra, S. M., A. T. Greer, J. W. Book, A. L. Deary, I. M. Soto, C. Culpepper, F. J. Hernandez, and **T. N. Miles** (2019), Acoustic detection of zooplankton diel vertical migration behaviors on the northern Gulf of Mexico shelf, *Limnol. Oceanogr.*, (Lampert 1989), lno.11171, doi:10.1002/lno.11171.
 35. Feehan, C. J., W. C. Sharp, **T. N. Miles**, M. S. Brown, and D. K. Adams (2019), Larval influx of *Diadema antillarum* to the Florida Keys linked to passage of a Tortugas Eddy, *Coral Reefs*, 38(2), 387–393, doi:10.1007/s00338-019-01786-9.
 36. Bushnell, M., S. Glenn, J. Kohut, J. McDonnell, **T. Miles**, G. Saba, and O. Schofield (2018), Workforce development supporting the blue economy: Using recent community-developed material in operational oceanography curricula: A tribute to Vembu Subramanian, *Mar. Technol. Soc. J.*, 52(3), 51–54, doi:10.4031/MTSJ.52.3.10.
 37. Greer, A.T., A.M. Shiller, E.E. Hofmann, J.D. Wiggert, S.J. Warner, S.M. Parra, C. Pan, J.W. Book, D. Joung, S. Dykstra, J.W. Krause, B. Dzwonkowski, I.M. Soto, M.K. Cambazoglu, A.L. Deary, C. Briseño-Avena, A.D. Boyette, J.A. Kastler, V. Sanial, L. Hode, U. Nwankwo, L.M. Chiaverano, S.J. O’Brien, P.J. Fitzpatrick, Y.H. Lau, M.S. Dinniman, K.M. Martin, P. Ho, A.K. Mojzis, S.D. Howden, F.J. Hernandez, I. Church, **T.N. Miles**, S. Sponaugle, J.N. Moum, R.A. Arnone, R.K. Cowen, G.A. Jacobs, O. Schofield, and W.M. Graham. (2018). Functioning of coastal river-dominated ecosystems and implications for oil spill response: From observations to mechanisms and models. *Oceanography* 31(3), <https://doi.org/10.5670/oceanog.2018.302>
 38. Zhang, F., M. Li, and **T. Miles** (2018), Generation of Near-Inertial Currents on the Mid-Atlantic Bight by Hurricane Arthur (2014), *J. Geophys. Res. Ocean.*, 123(4), 3100–3116, doi:10.1029/2017JC013584.

39. Ramos, A. G., Garcia-Garrido, V. J., Mancho, A. M., Wiggins, S., Coca, J., Glenn, S., Schofield, O., Kohut, J., Aragon, D., Kerfoot, J., Haskins, T., **Miles T. N.**, Haldeman, C., Strandkov, N., Allsup, B., Jones, C., Shapiro, J., (2018) Lagrangian coherent structure assisted path planning for transoceanic autonomous underwater vehicle missions. *Sci. Rep.*, 8(1), doi:10.1038/s41598-018-23028-8.
40. Schofield, O., Glenn, S., Kohut, J., **Miles, T.**, Roarty, H., Saba, G., McDonnell, J. (2017) Developing practical data skills in undergraduate students using ocean observatories. *Marine Technology Society* 52(1): 1-7.
41. **Miles, T.**, Seroka, G. and Glenn S., (2017), Coastal ocean circulation during Hurricane Sandy, *J. Geophys. Res. Ocean.*, 122(9), 7095–7114, doi:10.1002/2017JC013031
42. Goni, G.J., R.E. Todd, S.R. Jayne, G.R. Halliwell, S. Glenn, J. Dong, R. Curry, R. Domingues, F. Bringas, L. Centurioni, S.F. DiMarco, **T. Miles**, J. Morell, L. Pomales, H.-S. Kim, P.E. Robbins, G.G. Gawarkiewicz, J. Wilkin, J. Heiderich, B. Baltes, J.J. Cione, G. Seroka, K. Knee, and E.R. Sanabia. (2017) Autonomous and Lagrangian ocean observations for Atlantic tropical cyclone studies and forecasts. *Oceanography*, 30(2):85-95, doi:10.5670/oceanog.2017.227
43. Seroka, G., **T. Miles**, Y. Xu, J. Kohut, O. Schofield, and S. Glenn (2017), Rapid shelf-wide cooling response of a stratified coastal ocean to hurricanes, *JGR-Oceans: Special Section "Oceanic Responses and Feedbacks to Tropical Cyclones"*, doi:10.1002/2017JC012756
44. Schofield, O. et al. (2017), Decadal variability in coastal phytoplankton community composition in a changing West Antarctic Peninsula, *Deep Sea Res. Part I Oceanogr. Res. Pap.*, doi:10.1016/j.dsr.2017.04.014
45. **Miles, T.**, Lee, S.H., Wahlin, A., Schofield, O., Ha, H.K., Assman, K., (2016) Observations of the Dotson Ice Shelf outflow, *Deep Sea Research II* doi:10.1016/j.dsr2.2015.08.008
46. Seroka G., **Miles T.**, Xu, Y., Kohut, J., Schofield, O., Glenn, S., (2016) Hurricane Irene Sensitivity to Stratified Coastal Ocean Cooling *Mon. Wea. Rev.*, 0, doi: 10.1175/MWR-D-15-0452.1
47. Glenn, S., **Miles, T.**, Seroka., G.N., Xu, Y., Forney, R., Yu, F., Roarty, H., Schofield, O., Kohut, J., (2016) Stratified Coastal Ocean Interactions with Tropical Cyclones, *Nature Communications* 7
48. Schofield, O., **Miles, T.**, Alderkamp, A.C., Lee, S.H., Haskins, C., Rogalsky, E., Sipler, R., Sherrell, R., Yager, P., (2015) *In situ* phytoplankton distributions in the Amundsen Sea polynya measured by autonomous gliders, *Elementa*, 3:000073, doi:10.12952/journal.elementa.000073
49. **Miles, T.**, G. Seroka, J. Kohut, O. Schofield, and S. Glenn (2015), Glider observations and modeling of sediment transport in Hurricane Sandy, *J. Geophys. Res. Ocean.*, doi:10.1002/2014JC010474

50. Schofield, O., Jones, C., Kohut, J., Kremer, U., **Miles, T.**, Saba, G., Webb, D., Glenn, S., (2015) Developing Coordinated Communities of Autonomous Gliders for Sampling Coastal Ecosystems, *Marine Technology Society Journal*, 49, 3, doi: <http://dx.doi.org/10.4031/MTSJ.49.3.16>
51. Kohut, J., Bernard, K., Fraser, W., Oliver, M., Statscewich, H., Winsor, P., and **Miles, T.** (2014). Studying the impacts of local oceanographic processes on Adelie penguin foraging ecology. *Marine Technology Society Journal*, 48, 5 doi:10.4031/MTSJ.48.5.10
52. Saba, G. K., Fraser, W. R., Saba, V. S., Iannuzzi, R. A., Coleman, K. E., Doney, S. C., Ducklow, H. W., Martinson, D. G., **Miles, T. N.**, Patterson-Fraser, D. L., Stammerjohn, S. E., Steinberg, D. K., and Schofield, O. (2014) Winter and spring controls on the summer food web of the coastal West Antarctic Peninsula. *Nature Communications*. 5:4318 doi: 10.1038/ncomms5318
53. **Miles, T.**, Glenn, S., Schofield, O. (2013), Temporal and spatial variability in fall storm induced sediment resuspension on the Mid-Atlantic Bight. *Continental Shelf Research*, <http://dx.doi.org/10.1016/j.csr.2012.08.006>
54. **Miles, T.**, and He, R. (2010). Seasonal Surface Ocean Temporal and Spatial Variability of the South Atlantic Bight: Revisiting with MODIS SST and Chl-a imagery, *Continental Shelf Research*, V30, 1951-1962.
55. **Miles, T.**, He, R., M. Li (2009). Characterizing the South Atlantic Bight Seasonal Variability and Cold Water Event in 2003 Using a Daily Cloud-free SST and Chlorophyll Analysis. *Geophysical Research Letters*, 36, L02604, doi:10.1029/2008GL036393

Book Chapters

1. Ye, F., **Miles., T.**, Ezzat, A.A., (2023) Offshore Wind Energy Prediction Using Machine Learning with Multi-Resolution Inputs in “*Multimodal and Tensor Data Analytics for Industrial Systems*” Improvement Springer Optimization and Its Application (Accepted)
2. N., Beaird, S. Glenn, **T. Miles**, G. Saba, J. Kohut, O. Schofield (2021), Case study: RUCOOL Operational Oceanography Masters – Workforce development case study, in *Preparing a Workforce for the New Blue Economy*, by Richard Spinrad, pp 417-424, Elsevier.
3. Yi, X., S. Glenn, F. Carvalho, C. Jones, J. Kohut, J. McDonnell, **T. Miles**, G. Seroka, and O. Schofield (2019), Glider Technology Enabling a Diversity of Opportunities With Autonomous Ocean Sampling, in *Challenges and Innovations in Ocean In Situ Sensors*, pp. 367–374, Elsevier.
4. Schofield, O., Aragon, D., Jones, C., Kohut, J., **Miles, T.N.**, Roarty, H., Saba, G., Yi, X., Glenn, S. (2018). Maturing glider technology providing a modular platform capable of mapping ecosystems in the ocean. In: Delory, E., Pearlman, J., eds., *Challenges and Innovations in Ocean In-Situ Sensors*. Elsevier, Cambridge, MA. pp. 173-181.

Conference Proceedings (*student mentee)

1. M. Smith, S. Glenn, **T.N. Miles**, A.H. Knap, S.F. Dimarco, R.M. Jiménez, D. Salas de León, M.G. Reali, E. Escalante, "The Yucatan HF Radar Network: A Multinational Technology Demonstration to Advance Monitoring of Western Boundary Currents" Marine Technology Society Oceans 2025 conference, Chicago, October 2025.
2. *L. Hopson, **T.N. Miles**, J. Gradone, S. Glenn, D. Wilson, "Autonomous Underwater Glider Observations during Hurricane Beryl (2024)." Marine Technology Society Oceans 2025 conference, Chicago, October 2025.
3. J. Engdahl, D., Aragon, K., Coleman, **T.N., Miles**, G Saba, J Kohut, et al., "Risk Assessment of Underwater Glider Turbine Interaction" Marine Technology Society Oceans 2024, Halifax, Nova Scotia 2024.
4. E.M. Mule, et al., "Real-time probabilistic reachability forecasting for gliders in the Gulf of Mexico" Marine Technology Society Oceans 2024, Halifax, Nova Scotia 2024.
5. *S. Scopazzi, **T.N., Miles**, J., Gradone, D., Aragon, A., Lopez, "Sensor and Behavior Frameworks for implementing Backseat Driver on a Slocum Glider", Marine Technology Society Oceans 2024, Halifax, Nova Scotia 2024.
6. *L. Hopson, **T.N. Miles**, "Co-evolution of Air and Sea: How do Landfalling Tropical Cyclones Influence Coastal Ocean Circulation?" 36th Conference on Hurricanes and Tropical Meteorology, Long Beach California, May 2024.
7. S.F. DiMarco et al., "Applications of Adaptive Sampling Strategies of Autonomous Vehicles, Drifters, Floats, and HF-Radar, to Improve Loop Current System Dynamics Forecasts in the Deepwater Gulf of Mexico." Paper presented at the Offshore Technology Conference, Houston, Texas, May 2023. doi: <https://doi.org/10.4043/32459-MS>
8. *R. Horwitz, **T. N. Miles**, D. Munroe and J. Kohut, "Investigating the Overlap Between the Mid-Atlantic Bight Cold Pool and Offshore Wind Lease Areas," OCEANS 2022, Hampton Roads, Hampton Roads, VA, 2022, pages 1-6, doi: 10.1109/OCEANS47191.2022.9977050
9. *J.C. Gradone, E. J. Hunter, J. Mullison and **T. N. Miles**, "Development of Onboard Processing Capabilities for a Slocum Glider Acoustic Doppler Current Profiler," OCEANS 2021: San Diego – Porto, San Diego, CA, 2021, pages 1-5, doi: 10.23919/OCEANS44145.2021.9705895
10. J. Dicopoulos et al., "Weather Research and Forecasting model validation with NREL specifications over the New York/New Jersey Bight for offshore wind development," OCEANS 2021: San Diego – Porto, San Diego, CA, 2021, pages 1-7, doi: 10.23919/OCEANS44145.2021.9705742
11. *S. Coakley, **T.N. Miles**, S. Glenn and H. S. Lim, "Observation-Large eddy simulation comparison of ocean mixing under Typhoon Soulik (2018)," OCEANS 2021: San Diego – Porto, San Diego, CA, 2021, pages 1-7, doi: 10.23919/OCEANS44145.2021.9705670
12. *J. Anarumo, **T. Miles**, H. Roarty, J. Kohut and N. Beaird, "An Open-Source Software Application for Drifter Trajectory Prediction in the Mid-Atlantic Bight," Global Oceans

- 2020: Singapore – U.S. Gulf Coast, Biloxi, MS, 2020, pages 1-8, doi: 10.1109/IEEECONF38699.2020.9389124
13. M. F. Aristizabal-Vargas, **T. Miles**, S. Glenn, H. -S. Kim and A. Mehra, "Validation of the HWRF-POM and HWRF-HYCOM hurricane forecasting systems during Hurricane Dorian using glider observations," Global Oceans 2020: Singapore – U.S. Gulf Coast, Biloxi, MS, 2020, pages 1-5, doi: 10.1109/IEEECONF38699.2020.9389066
 14. Vargas, M. A., **Miles, T.N.**, Glenn, S., Hogan, P., Douglas Wilson, W. Watlington, R., and Lacour, B., 2019, "Impact of Glider Data Assimilation on the Global Ocean Forecasting System during the 2018 Hurricane Season", OCEANS 2019 Marine Technology Society -Seattle, Seattle, WA, pages 1–5, doi.org/10.23919/OCEANS40490.2019.8962824
 15. Saba, G. K., *Wright-Fairbanks, E., **Miles, T.N.**, Chen, B., Cai, W.-J., Wang, K., Barnard, A.H., Branham, C.W, and Jones, C.P., 2018, "Developing a profiling glider pH sensor for high resolution coastal ocean acidification monitoring", OCEANS 2018 Marine Technology Society -Charleston, Charleston, SC, pages 1–8, doi.org/10.1109/OCEANS.2018.8604918
 16. Kohut, J., Glenn, S., McDonnell, J., **Miles, T.N.**, Saba, G., and Schofield, O., (2018), "Workforce Development Supporting the Blue Economy: A Master's Program of Integrated Ocean Observing at Rutgers University", OCEANS 2018 Marine Technology Society -Charleston, Charleston, SC, pages 1–5, doi.org/10.1109/OCEANS.2018.8604625
 17. **Miles, T.N.**, Kohut, J., Slade, W. and Gong, D. (2018), "Suspended particle characteristics from a glider integrated LISST sensor", OCEANS 2018 Marine Technology Society- Charleston, Charleston, SC, pages 1–5, doi.org/10.1109/OCEANS.2018.8604627
 18. *Gonzalez-Morabito, C., Aragon, D., **Miles, T.N.**, Kohut, J., and Glenn, S., (2018), "Flight dynamics of Slocum gliders in intermediate-water hurricane: Rutgers University Center for Ocean Observing Leadership (RU COOL)", OCEANS 2017 Marine Technology Society Anchorage, Anchorage, Alaska, pages 1-6, <https://ieeexplore.ieee.org/document/8232227>
 19. Schofield, O., Brown, M., Carvahlo, F. Couto, N., Crowley, M., **Miles, T.N.**, Nardelli, S., Waite, N. and Saba, G. (2017), "Decadal trends in phytoplankton communities along a rapidly changing West Antarctic Peninsula", OCEANS 2017- Anchorage, Anchorage, Alaska, pages 1–6, <https://ieeexplore.ieee.org/document/8232269>
 20. Glenn, S., Schofield, O., Kohut, J., McDonnell, J., **Miles, T.N.**, Saba, G., Roarty, H., and Crowley, M. (2017), "Workforce development for the blue economy: Rutgers masters in integrated ocean observing", OCEANS 2017 - Anchorage, Anchorage, Alaska, pages 1–4, <https://ieeexplore.ieee.org/document/8232336>
 21. Glenn, S., **Miles, T.N.**, Seroka, G., Forney, R., Roarty, H., Schofield, O., Kohut, J., Xu, Y., and Yu, F., (2016), "Stratified coastal ocean processes in hurricanes and typhoons enhance ahead-of-eye cooling and reduce storm intensity", OCEANS 2016 – Shanghai, Shanghai, China, pages 1-6, doi.org/10.1109/OCEANSAP.2016.7485440

22. Glenn, S., Schofield, O., Saba, G., **Miles, T.N.**, Roarty, H., and Kohut, J. (2015), “The ocean is our classroom: A 4-year research track for undergraduate exploration, research and discovery in oceanography”, OCEANS 2015 – Marine Technology Society - Washington, Washington, DC, pages 1–5, doi.org/10.23919/OCEANS.2015.7404567
23. Aragon, D., Glenn, S., **Miles, T.N.**, and Curry, R., (2015), “Glider performance during Hurricane Gonzalo”, OCEANS 2015 – Marine Technology Society- Washington, Washington, DC, pages 1–5, doi.org/10.23919/OCEANS.2015.7404371
24. **Miles, T.N.**, Seroka, G., and Glenn, S. (2015), “Coastal ocean mixing and advection during Hurricane Sandy and Arthur”, OCEANS 2015 – Marine Technology Society- Washington, Washington, DC, pages 1–7, doi.org/10.23919/OCEANS.2015.7404404
25. Schofield, O., Kohut, J., Kremer, U., **Miles, T.N.**, Saba, G., Glenn, S. Jones, C., and Webb, D. (2015), “Gliders as maturing technology: Using gliderpalooza as means to develop an integrated glider community”, OCEANS 2015 – Marine Technology Society- Washington, Washington, DC, pages 1–5, doi.org/10.23919/OCEANS.2015.7404588
26. *Ramsay, L., Morrison, A., Kohut, J., Schofield, O., **Miles, T.N.**, Glenn, S., and Lintz, C. (2015), “Glider advancements in efficiency: Enhancing factors necessary for ocean-wide flights”, OCEANS 2015 – Marine Technology Society- Washington, Washington, DC, pages 1–5, doi.org/10.23919/OCEANS.2015.7404553
27. Seroka, G., **Miles, T.N.**, Dunk, R., Kohut, J., Glenn, S., and Fredj, E. (2015), “Sea breeze, coastal upwelling modeling to support offshore wind energy planning and operations”, OCEANS 2015 – Marine Technology Society Washington, Washington, DC, pages 1–6, doi.org/10.23919/OCEANS.2015.7404456
28. *Morrison, A. et al, (2015), “Model comparison for transatlantic ocean glider flight: Student analysis of modern circumnavigation”, OCEANS 2015 – Marine Technology Society Washington, Washington, DC, pages 1–7, doi.org/10.23919/OCEANS.2015.7404505
29. **Miles, T.N.**, Glenn, S., Schofield, O., Kohut, J. and Seroka, G. (2014), “Sediment transport in Hurricane Sandy”, 2014 OCEANS- St. John’s, St. John’s, NL, Canada, pages 1–8, doi:10.1109/OCEANS-TAIPEI.2014.6964331
30. Glenn, S., Seroka, G., **Miles, T.N.**, Xu, Y., Roarty, H., Kohut, J., and Schofield, O., (2014), “The role of regional-scale ocean observations for improved hurricane intensity and impact forecasts in coastal regions”, OCEANS 2014 - TAIPEI, Taipei, Taiwan, pages 1–9, doi.org/10.1109/OCEANS-TAIPEI.2014.6964331
31. **Miles, T.N.**, Glenn, S., Kohut, J., Seroka, G., and Xu, Y. (2013), “Observations of Hurricane Sandy from a glider mounted aquadopp profiler”, 2013 OCEANS-San Diego, San Diego, CA, pages 1–8, doi:10.23919/OCEANS.2013.6741176
32. Glenn, S. et al, (2013), “Process-driven improvements to hurricane intensity and storm surge forecasts in the mid-atlantic bight: Lessons learned from hurricanes Irene and sandy, 2013 OCEANS, Marine Technology Society - Bergen, Bergen, Norway, pages 1–9, doi.org/10.1109/OCEANS-Bergen.2013.6608108

Invited Professional Meetings and Abstracts

1. “Gliders in the Gulf: Past, Present, and Future” Travis Miles and Barbara Kirkpatrick at the Underwater Glider User Group (UG2) workshop (May 19th 2026)
2. “Expanding ocean observing systems for weather and climate: From the Northeast US to the Caribbean Sea” University of Rhode Island Graduate School of Oceanography Physical Oceanography Seminar (October 31st 2025)
3. “Riders on the storm: 15 years of progress in glider hurricane sampling,” Teledyne Webb Research, Slocum Gliders User Conference (Oct 8, 2025)
4. “Offshore Wind and the Mid Atlantic Bight Cold Pool,” Old Dominion University Center for Coastal Physical Oceanography (CCPO) Seminar (Oct, 21, 2024)
5. “A Decade of Autonomous Underwater Observations of Tropical Cyclones: Co-evolution of the Ocean and Atmosphere,” KIOST JPA Meeting, Busan, Korea (Nov 13th 2023)
6. “A Decade of Autonomous Underwater Observations of Tropical Cyclones: Co-evolution of the Ocean and Atmosphere,” Tropical Cyclone and Ocean Interactions, Jeju Island, Korea (Nov 6th 2023)
7. “Autonomous underwater gliders for hurricane research and operations,” Mote Marine Laboratory, Sarasota FL (July 21st 2023)
8. “Renewable Energy in the MACRUC region” Panelist Mid-Atlantic Conference of Regulatory Utilities Commissioners, Nemaquin, PA (June 26th 2023)
9. “Ocean Gliders Impacts to Forecasting Suites” The Underwater Glider User Group (UG2), quarterly meeting, webinar (February 17th 2022)
10. “Hurricane forecasting and ocean robots: How new technology is changing the way we study and predict extreme storms” The Ronald E. Hatcher Science on Saturday Series, Princeton Plasma Physics Laboratory, Princeton NJ (March 05, 2022)
11. “An Oceanographers Overview of Offshore Wind” Princeton Agricultural Association Princeton NJ (August 25th 2022)
12. “Hurricanes and the Mid Atlantic Cold Pool: How Our Regional Ocean Can Protect Coastal Communities,” Rutgers University Marine Extension Program Seminar Series (May 24, 2021)
13. “Sediment Resuspension Observations from a Glider Integrated Sequoia Scientific LISST Particle Analyzer,” The Oceanography Society Webinar Series on Exploring Ocean Instrumentation (July 21st 2021)
14. “Marine Technology Society Townhall on Offshore Wind” Panelist, Oceans Conference, San Diego California (September 22, 2021)

15. “An Oceanographers Overview of Offshore Win and the Mid-Atlantic Bight Pool” Stockton University for the New Jersey Environmental Lobby, Galloway, NJ, (October 5, 2021)
16. “Autonomous underwater gliders for coastal hurricane operations and long-term climate monitoring,” Coastal and Estuarine Research Foundation 2021 Biennial Conference (November 3rd 2021)
17. “Empirical Corrections to Thermal Lag on Ocean Gliders”, RBR Technical Workshop, San Diego California, (February 17, 2020)
18. “Hurricane Gliders” The Underwater Glider User Group, UG2 Welcome Webinar, virtual, (December 9, 2020)
19. “Autonomous Underwater Gliders: Sensing capabilities and applications”, Korea Institute of Ocean Science and Technology (KIOST), National Oceanic and Atmospheric Administration -Korea Institute Ocean Science & Technology (NOAA-KIOST) Joint Program Agreement Meeting, Busan, South Korea, with Scott Glenn, (June 4-5, 2019)
20. “Stratified Coastal Ocean Interactions with Tropical Cyclones”, Stevens Institute of Technology, Hoboken, NJ, (April 16, 2019)
21. “Gliders, Observing Systems, and Numerical Modeling: Stratified Coastal Ocean Interactions with Tropical Cyclones.” World Meteorological Organization/Intergovernmental Oceanographic Commission (WMO/IOC) Data Buoy Cooperation Panel (DBCP) Third Pacific Islands Training Workshop on Ocean Observations and Data Applications, National Center for Ocean Standards and Meteorology, Hai Kou, China, (July 2018)
22. “Stratified Coastal Ocean Interactions with Tropical Cyclones: Coastal ocean circulation during Hurricane Sandy.” Fifth Capacity Building Workshop of the World Meteorological Organization/Intergovernmental Oceanographic Commission (WMO/IOC) Data Buoy Cooperation Panel (DBCP) for the North Pacific Ocean and Its Marginal Seas (NPOMS-5), Tianjin, China, with Scott Glenn, (June 4-7, 2017)
23. “Stratified coastal ocean interactions with tropical cyclones.” University of Maryland Center for Environmental Studies Horn Point Laboratory Seminar, Horn Point, Maryland, (February 2017)
24. “Glider applications for ocean observation networks and process-oriented numerical modeling.” Workshop on application of glider technology for sustained oceanographic observations under climate variability and climate change, Lima Peru, (2017)
25. “Stratified coastal ocean interactions with tropical cyclones.” Lamont-Doherty Earth Observatory Columbia University Physical Oceanography Seminar, Columbia University, New York, (May 2016)
26. “Observations of Hurricane Sandy from a Glider Mounted Current Profiler.” Fourth Capacity Building Workshop of the World Meteorological Organization/Intergovernmental Oceanographic Commission (WMO/IOC) Data Buoy

- Cooperation Panel (DBCP) for the North Pacific Ocean and Its Marginal Seas (NPOMS-4), Korea Maritime and Ocean University, Busan, South Korea, (November 2015)
27. “Glider Observations along the Dotson Ice Shelf.” 20th International Symposium on Polar Sciences, Korean Polar Research Institute, Incheon, South Korea, (May 2014)
 28. “Observations of Hurricane Sandy from a Glider Mounted Aquadop Profiler.” Nortek Technical Symposium, San Diego, CA, (September 2013)

Presentations at Professional Meetings and Conferences (*student mentee an)

1. *Hopson, L.N., **Miles, T.N.**, Gradone, J., Wilson, D., Looney, L.B., Glenn, S., “Autonomous Underwater Glider Observations in Category 5 Hurricane Beryl.” 105th Annual AMS Meeting 2025, New Orleans, LA, January 2025, paper ID 453574
2. Steinbarger, C., Jelenak, Z., Looney, L.B., **Miles, T.N.**, Zhang, J., “Coordinated Hurricane Observing through Collaborative Research Partnerships” AGU Fall Meeting 2024, Washington D.C., December 2024.
3. **Miles, T.N.**, Hopson, L.N., Gradone, J., Wilson, D., Looney, L.B., Glenn, S., “Autonomous Underwater Glider Observations in Category 5 Hurricane Beryl.” AGU Fall Meeting 2024, Washington D.C., December 2024.
4. **Miles, T.N.**, Coakley, S., Engdahl J., Rudzin J., Tsei, S, and S. Glenn. “Ocean mixing during Hurricane Ida (2021): The impact of a freshwater barrier layer” AGU Ocean Sciences Meeting, 2024.
5. *Gradone, J., Wilson, D., Glenn, S., **Miles, T.N.**, “Observations of Water Mass Properties in the Caribbean Through-Flow and their Implications for the Atlantic Meridional Overturning Circulation” AGU Ocean Sciences Meeting, 2024.
6. *Horwitz, B., **Miles, T.N.**, “Overlap between the Mid-Atlantic Bight Cold Pool and Offshore Wind Energy Areas” AGU Ocean Sciences Meeting, 2024.
7. *R. Cassin, **Miles, T.N.**, Pareja-Roman, L.F., “Investigating Potential Impacts of Offshore Wind Turbine Foundations on the Mid-Atlantic Bight Cold Pool” AGU Ocean Sciences Meeting, 2024. (Poster Presentation)
8. **Miles, T.N.**, Coakley, S., Engdahl J., Rudzin J., Tsei, S, and S. Glenn. Ocean mixing during Hurricane Ida (2021): the impact of a freshwater barrier layer. ASLO Ocean Sciences Meeting, 2023. (Poster Presentation)
9. *Gradone, J.C., **Miles, T.**, Glenn, S., & Wilson, D. Searching for Pathways of Cross-Equatorial Water Mass Transport in the Eastern Caribbean. ASLO Ocean Sciences Meeting, 2023. (Poster Presentation)
10. Bailey, K., Stienbarger, C., **Miles, T. N.**, & Rudzin, J. E. (2022, December). Advances in Hurricane Intensity Forecasting and Research Through the Use of Innovative Ocean Observing Technologies I Oral. In Fall Meeting 2022. AGU.

11. *Gradone, J.C., **Miles, T.**, Glenn, S., & Wilson, D. Slocum Glider ADP Based Observations of
12. Caribbean Through-Flow and Their Implications for Global Climate. Underwater Glider User Group (UG2) Meeting. Seattle, WA. September 21, 2022.
13. *Horwitz, B., **Miles, T.N.**, Munroe, D., and Kohut, J., Investigating the Overlap between the Mid-Atlantic Bight Cold Pool and Offshore Wind Lease Areas (2022, October). Marine Technology Society Oceans 2022
14. *Gradone, J.C., **Miles, T.**, Glenn, S., & Wilson, D. Observing Essential Ocean Features in the Eastern Caribbean for a Safe and Predicted Ocean. ASLO Ocean Sciences Meeting, 2022. (Oral **Presentation**)
15. Stienbarger, C., Bailey, K., Goni, G. J., Kim, H.-S., **Miles, T. N.**, and Rudzin, J. E., “NOAA Works Toward Improving Tropical Cyclone Intensity Forecasts Through Coordinated Ocean Observing”, vol. 2022, 2022.
16. *Tsei, S., Howden, S. D., Zhang, J., and **Miles, T. N.**, “Hurricane Ida (2021): Coastal Ocean Thermal Structure and Rapid Intensification over the Louisiana Shelf Region”, vol. 2022, 2022.
17. Bourrin, F., Gentil, M., & **Miles, T.** (2022, May). Suspended Particle Characteristics from Glider Observations in a Region of Freshwater Influence. In EGU General Assembly Conference Abstracts (pp. EGU22-5170).
18. Le Henaff, M., Goni, G., Bringas, F., Chardon-Maldonado, P., Morell, J., Kim, H.S., Trinanes, J., Zhang, J., Glenn, S., **Miles, T.** and Aristizabal, M., 2021, December. Hurricane glider operations in the Caribbean region for hurricane research and forecasts. In AGU Fall Meeting Abstracts (Vol. 2021, pp. OS25C-1023)
19. *Coakley S, Stienbarger C, Goni G, Kim HS, Le Henaff M, Lumpkin R, **Miles T.**, Smith E, Thurston S. Integrating Ocean Observations to Improve Hurricane Intensity Forecasts. In AGU Fall Meeting Abstracts 2021 Dec (Vol. 2021, pp. OS25C-1022).
20. *Wright-Fairbanks, E.K., **Miles, T.**, Chen, B., Cai, W.-J., Saba, G. 2021. Glider-based observations reveal seasonal pH and aragonite saturation state variability in coastal U.S. Mid-Atlantic shellfishery management zones. Talk presented at the National Shellfisheries Association annual meeting (virtual due to COVID). March 2021.
21. *Smith, H. et al., “Robotic Ocean Gliders as Vicarious Ocean Color Calibration and Validation data for MODIS-Aqua, Suomi-NPP VIIRS, JPSS-1, and PACE.”, vol. 2020, 2020.
22. *Proszowski A, **Miles TN**, Wang RQ, Brodie JF. Assessing the Impact of Spatial Variability and Wake Effects on Power Prediction for NJ Offshore Wind Energy Area. In AGU Fall Meeting Abstracts 2020 Dec (Vol. 2020, pp. GC125-04).
23. **Miles, T.N.**, Arango, H., and Wilkin, J., “A Coupled Atmosphere-Ocean Modeling Framework for the Regional Ocean Modeling System,” 1st UFS Users Workshop webinar, (July 28, 2020)

24. Glenn, S., **Miles, T.N.**, Seroka, G., *Watkins, C., Aristizabal, M., Wilson, D., Watlington, R., Coakley, S., Ramos-Valle, A., Lim, H.S., Xu, Y., “Rapid Co-evolution of the ocean and atmosphere in tropical cyclones and typhoons”, Global Ocean Monitoring Hurricane Steering Committee, webinar, (June 2, 2020) Miles, T.N., Mid Atlantic Bight: Summary of 2019 Glider Operations, Hurricane Gliders Workshop, webinar (April 28, 2020)
25. **Miles, T.N.**, “Planned 2020 Hurricane Glider Operations”, Hurricane Glider Workshop, webinar, (April 28, 2020)
26. Aristizabal, M.F., Kim, H.S., **Miles, T.N.**, Glenn, S., Mehra, A., “Evaluation of the Ocean Initial Conditions and Evolution of the Ocean Mixed Layer Temperature on The Weather Research and Forecast Modeling System for Hurricanes/The Princeton Ocean Model for Tropical Cyclones (HWRF-POM) Forecasting Model during Hurricane Dorian”, American Geophysical Union (AGU) Ocean Sciences Meeting, San Diego California, (February 16-21 2020)
27. Wang, R.Q., **Miles, T.N.**, Brodie, J.F., “Multi-scale Interaction between Wind Turbines and Coastal Processes: coupling OpenFAST with a regional coupled air-sea modeling system”, American Geophysical Union (AGU) Ocean Sciences Meeting, San Diego California, (February 16-21, 2020)
28. Brodie, J.F., Frei, B.P., **Miles, T.N.**, Veron, D.E., Allen, E., “Investigating the Role of Coastal Upwelling and Synoptic Conditions in the Prediction of Wind Ramp Events”, American Geophysical Union (AGU) Ocean Sciences Meeting, San Diego California, (February 16-21, 2020)
29. *Murphy, S., **Miles, T.N.**, Brodie, J.F., Crowley, M.F., Oliver, M.J., Nazarro, L.J., “Impact of Rapidly Evolving SST fields during Coastal Upwelling Events on Offshore Wind Power Production”, American Geophysical Union (AGU) Ocean Sciences Meeting, San Diego California, (February 16-21, 2020)
30. *Wright-Fairbanks, E., Saba, G., Chen, B. Cai, W.J., **Miles, T.N.**, “Glider-Based Observations Reveal Seasonal pH and Aragonite Saturation State Variability in Coastal US Mid-Atlantic Shellfishery Management Zones,” American Geophysical Union (AGU) Ocean Sciences Meeting, San Diego California, (February 16-21, 2020)
31. *Wang, H., Gong, D., Yu, H.C., Zhang, Y.J., Harris, C.K., Friedrichs, M.A.M., **Miles, T.N.**, Hudson, J., “Quantifying Shelf-Basin Exchange at Submarine Canyons in the southern Middle Atlantic Bight using data from Underwater Glider Surveys and Numerical Modeling Experiments”, American Geophysical Union (AGU) Ocean Sciences Meeting, San Diego California, (February 16-21, 2020)
32. Aristizabal, M.F., **Miles, T.N.**, Glenn, S.M., Wilson, W.D., Watlington, R., “Mixed Layer Temperature Variability During the 2018 Hurricane Season in the Caribbean Ocean”, American Geophysical Union (AGU) Ocean Sciences Meeting, San Diego California, (February 16-21, 2020)
33. Glenn, S.M., **Miles, T.N.**, Aristizabal, M.F., Watkins, C. Him, H.S., “Scientific Results from the 2018 & 2019 North Atlantic Hurricane Glider Picket Lines”, American

- Geophysical Union (AGU) Ocean Sciences Meeting, San Diego California, (February 16-21, 2020)
34. **Miles, T.N.**, Valle, A.R., Brodie, J.F., Seroka, G.N., Coakley, S., “Sensitivity of storm surge to stratified coastal ocean cooling in Hurricane Irene”, American Geophysical Union (AGU) Ocean Sciences Meeting, San Diego California, (February 16-21, 2020)
 35. *Wright-Fairbanks, E. Saba, G.K., Chen, B., Cai, W.-J., Wang, K., Bernard, A.H., Branham, C.W., Jones, C.P., and **Miles, T.N.** Glider-based observations reveal seasonal pH and aragonite saturation state variability in coastal U.S. Mid-Atlantic shellfishery zones. Poster presented at the Ocean Acidification Community Meeting, Miami, FL., January 2020.
 36. **Miles, T.N.**, and Brodie, J.F, “New Jersey Offshore Wind Resource Modeling: The Power of Partnerships”, Offshore Wind International Partnering Forum, New York, New York, (2019)
 37. **Miles, T.N.**, Arango, H., and Wilkin, J.,” Air-sea coupling with the Earth System Modeling Framework/The National Unified Operational Prediction Capability (ESMF/NUOPC) layer”, Weather Research and Forecasting (WRF) and Regional Ocean Modeling System (ROMS), U.S.(Integrated Ocean Observing System)” IOOS 2019 Coastal and Ocean Modeling Testbed (COMT) Annual Meeting, Silver Spring, MD, (2019)
 38. **Miles, T.N.**, Slade, W., Kohut, J., and Gong, D., “Suspended particle characteristics from a glider integrated LISST sensor”, Underwater Gliders User Group/ Everyone’s Gliding Observatories (EGO) Meeting, Rutgers University, (2019)
 39. Gouldman, C., **Miles, T.N.**, Edwards, C., “Under Florence’s Wake: Gliders, Hurricanes, and Ocean Observing”, Oceans 2018 Marine Technology Society, Charleston, SC, (2018)
 40. **Miles, T.N.**, Glenn, S., Seroka, G., “Coastal Ocean Circulation During Hurricane Sandy”, American Geophysical Union (AGU) Ocean Sciences Meeting, Portland, OR, (2018)
 41. **Miles, T.N.**, Slade, W., Kohut, J., and Gong, D., “Suspended particle characteristics from a glider integrated LISST sensor”, Oceans 2018 Marine Technology Society (MTS), Charleston, SC, (2018)
 42. **Miles, T.N. et al.**, “Initial impacts of the Hurricane Sentinel glider fleet deployed during the 2018 hurricane season”, American Geophysical Union (AGU) Fall Meeting, Washington, DC, (2018)
 43. **Miles, T.N.**, “Stratified coastal ocean interactions with tropical cyclones”, Mid Atlantic Bight Physical Oceanography Meeting (MABPOM), (2016)
 44. **Miles, T.N.**, Glenn, S., Schofield, O., Kohut, J., Seroka, G., “Sediment transport in Hurricane Sandy,” Glider Observations and Regional Ocean Modeling. 7 th Everyone’s Gliding Observatories (EGO) Conference (2016)

45. **Miles, T.N.**, Arnone, B., Howden, S., Jones, C., Martin, K., Vandermeulen, R., Decolibus, C., Schofield, O., “Enabling Slocum Glider Flight on a River-Dominated Continental Shelf”, Gulf of Mexico Research Initiative Oil Spill Conference (2016)
46. **Miles T.N.**, Glenn, S., Schofield, O., Kohut, J., Seroka, G., “Sediment transport in Hurricane Sandy”, Glider Observations and Regional Ocean Modeling. 7th Everyone’s Gliding Observatories (EGO) Conference (2016)
47. **Miles, T.N.**, Glenn, S., Kohut, J., Schofield, O., Seroka, G., “Sediment resuspension and transport in Hurricane Sandy: Comparisons of glider observations with model results”, American Geophysical Union (AGU) Ocean Sciences Meeting, Honolulu, HI, (2014)
48. **Miles, T.N.**, Glenn, S., Schofield, O., Kohut, J., Seroka, G., “Sediment transport in Hurricane Sandy”, Oceans 2014 Marine Technology Society, St. Johns, Newfoundland, Canada, (2014)
49. **Miles, T.N.**, Schofield, O., Martinson D., “Coastal ocean mixing near Palmer Station Antarctica: Observations from a glider mounted Acoustic Doppler Current Profiler (ADCP)”, Association for the Sciences of Limnology and Oceanography (ASLO) Ocean Sciences Meeting, New Orleans, LA, (2013)
50. **Miles, T.N.**, Glenn, S., Schofield, O., Kohut, J., Seroka, G., Xu, Y., “Observations of Hurricane Sandy from a Glider Mounted Aquadopp Profiler”, Nortek Technical Symposium, San Diego, CA, 2013 (Invited) and Oceans 2013 Marine Technology Society, San Diego, CA, (2013)
51. Glenn, S., Schofield, O., Kohut, J., Bowers, L., Crowley, M., Dunk, R., Kerfoot, J., **Miles, T.N.**, Palamara, L., Roarty, H., Seroka, G., Yi X., Titlow, J., Brown, W., Boicourt, W., Atkinson, L., Seim, H., “(2012) Impact of ocean observations on hurricane forecasts in the Mid-Atlantic: Forecasting lessons learned from Hurricane Irene”, Marine Technology Society Oceans, (2012)
52. **Miles, T.N.**, Schofield, O., Martinson, D., “Physical forcing of a Western Antarctic Peninsula ecosystem: observations from a coastal ocean observing network at Palmer Station. Long Term Ecological Research (LTER)”, All Sciences Meeting, Estes Park, Colorado, (2012)
53. Glenn, S., Roarty, H., Kohut, J., **Miles, T.N.**, Kerfoot, J., Schofield, O., Brown, W., Atkinson, L., Boicourt, B., “Observations of Storm Response and Sediment Transport on the Middle Atlantic Bight Continental Shelf,” American Geophysical Union (AGU) Ocean Sciences Meeting, Portland, OR, (2010)
54. **Miles, T.N.**, and He, R., “Surface Ocean Temporal and Spatial Variability of the South Atlantic Bight: Revisiting with Cloud-free Reconstructions of Moderate Resolution Imaging Spectroradiometer (MODIS) Concurrent Self-System Therapy (SST) and Chl-a Imagery”, American Geophysical Union (AGU) Ocean Sciences Meeting, Portland, OR, (2010)
55. **Miles, T.N.**, and He, R., “Covariations of satellite sea surface temperature and ocean color in the South Atlantic Bight”, Ocean Sciences Meeting, Orlando, FL, (2008)

56. Other Presentations, Lectures, and Demonstrations

57. **Miles, T.N.**, Goni, G., and Glenn, S., “Hurricane Gliders: New Observing Technologies and Concept of Operations document (CONOPS)”, National Oceanic and Atmospheric Administration (NOAA), webinar, (Sept 3, 2020)
58. Edwards, C. and **Miles T.N.**, Hurricane Gliders: Improving Tropical Storm Intensity Forecasts with Real Time Data edited, The Southeast Coastal Ocean Observing Regional Association (SECOORA), webinar, (May 5, 2020)
59. **Miles, T.N.**, “Hurricanes and Robots: How new technology is changing the way we predict and study extreme storms,” National Oceanic and Atmospheric Administration (NOAA) Planet Stewards, (May 2019)
60. Glenn, S., Seroka, G., **Miles, T.N.**, (presenter), Xu, Y., Roarty H., Kohut J., Schofield, O., “U.S. Integrated Ocean Observing System (IOOS) Responds to Hurricanes Irene and Sandy”, Teledyne-Webb Research, Falmouth, MA, (2013)
61. **Miles, T.N.**, and Schofield, O., “20 Years of Oceanographic and Atmospheric Data at Palmer Station Antarctica, Long Term Ecological Research (LTER) Site Review”, Palmer Station, Antarctica, (2011)

Chaired Conference Sessions

1. TC-Ocean interactions: From observations to models, AGU 2024 Ocean Sciences Meeting (February 18th, 2024)
2. Pairing autonomous monitoring with modeling to expand capacity and develop new understanding of coastal ocean systems, ASLO 2022 Ocean Sciences Meeting (February 24th 2022)
3. Advances in Hurricane Intensity Forecasting and Research Through the Use of Innovative Ocean Observing Technologies, AGU 2022 Fall Meeting (December 13, 2022)

Awards

2025 SEBS Rutgers Early Career International Excellence Award	2025
The Oceanography Society Ocean Observing Team Award	2024
Bristlemouth Pioneer Program	2024
RBR 2020 Cohort Fellow	2020
Marine Technology Society Young Professional Award	2018
Fulbright Scholar - University of Gothenburg Sweden	2017
Teledyne-Webb Graduate Research Fellowship – Rutgers	2012 - 2014
NortekUSA Student Equipment Loan and Travel Grant – Rutgers	2012
Graduate School of New Brunswick Travel Award – Rutgers	2010
North Carolina NASA Space Grant Graduate Fellowship – NCSU	2008
Undergraduate Research Award – NCSU	2007
Marine Earth & Atmospheric Sciences Outstanding Senior – NCSU	2007
NSF Undergraduate Research Fellowship – Mote Marine Lab	2006
Park Scholarship (Full 4-year Merit Scholarship) – NCSU	2003 - 2007

Teaching Experience

Operational Oceanography Software Bootcamp – Rutgers	2025
Ocean Methods and Data Analysis – Rutgers	2025
Operational Oceanography Software Bootcamp – Rutgers	2024
Ocean Methods and Data Analysis – Rutgers	2024
Operational Oceanography Software Bootcamp – Rutgers	2023
Ocean Methods and Data Analysis – Rutgers	2023
Integrated Ocean Observing I – Rutgers	2022
Operational Oceanography Software Bootcamp – Rutgers	2022
Ocean Methods and Data Analysis – Rutgers	2022
Integrated Ocean Observing Field Methods II – Rutgers	2022
Integrated Ocean Observing II – Rutgers	2022
Integrated Ocean Observing I – Rutgers	2021
Operational Ocean Modeling and Visualization – Rutgers	2021
Operational Oceanography Software Bootcamp – Rutgers	2021
Integrated Ocean Observing Field Methods I – Rutgers	2021
Ocean Methods and Data Analysis – Rutgers (Canceled due to Covid)	2021
Integrated Ocean Observing II – Rutgers	2021
Operational Ocean Modeling and Visualization – Rutgers	2020
Ocean Methods and Data Analysis – Rutgers	2020
Operational Ocean Modeling and Visualization – Rutgers	2019
Ocean Methods and Data Analysis – Rutgers	2019
Physical Oceanography	2016
Ocean Methods and Data Analysis – Rutgers	2015
Teaching Assistant: Ocean Methods and Data Analysis – Rutgers	2013
Teaching Aid: Introduction to Oceanography – Rutgers	2010
Teaching Assistant: Introduction to Oceanography Lab – NCSU	2008
Teaching Assistant: Earth System Sciences –NCSU	2007

Workshops and Short Courses

Slocum Glider User Workshop – presenter	2025
Rockland Scientific Ocean Microstructure Glider training – host	2025
Underwater Glider User Group Bi-annual Meeting	2024
Understanding Gulf Ocean Systems Annual Meeting	2023
Global Ocean Observing System UN Co-Design Meeting – attendee	2023
Rutgers Offshore Wind Symposium	2023
UN Ocean Decade Tropical Cyclone Exemplar Workshop	2022
Rutgers Partners in Science Cold Pool Workshop – invited presenter	2019
NOAA-KIOST Ocean Research Panel Workshop – invited presenter	2019
IOOS-OAR Workshop	2019
Task Force Ocean workshops (3 sessions)	2017
NPOMS-6 Capacity Building Workshop of the WMO/IOC Haikou, China (invited)	2017
NPOMS-5 Capacity Building Workshop of the WMO/IOC Tianjin, China (invited)	2017
NPOMS-4 Capacity Building Workshop of the WMO/IOC Busan, Korea (invited)	2015
Pan-American Advanced Studies Institute (PASI): Amazon River-to-Shelf (selected)	2013
Alpine Summer School: Buoyancy Driven Flows (selected)	2010
Teledyne-Webb Slocum AUV training course	2007

Outreach Activities

The Challenger Mission: Capacity building in the Caribbean Sea (Barbados)	2025
Earth Day Every Day Webinar Series Marine Extension Webinar	2025

Mid-Atlantic Conference of Regulatory Utilities Commissioners Panelist	2023
The Ronald E. Hatcher Sci. on Saturday Lecture (Princeton Plasma Physics Lab)	2022
Governors STEM Scholars Panelist	2022
Panelist on Marine Technology and Offshore wind – MTS Oceans 2021	2021
New Jersey Environmental Lobby - Offshore wind and the Cold Pool	2021
Commerce Secretary Gina Ramando tour and panel on offshore wind	2021
NJDEP Commissioner LaTourette tour of RUCOOL	2021
Introductory Fisheries Science for Stakeholders Guest Lecture	2020
Partners in Science Workshop: Offshore wind and the MAB Cold Pool	2019
NOAA Planet Stewards Webinar	2019
GliderPalooza co-lead	2015 - 2016
NCSU Park Scholarship Alumni Mentor	2013 - present
NCSU Park Scholarship Interviewer	2011 - present
Volunteer Science Q&A at Palmer Station, Antarctica	2010 - 2012
Presenter Science Saturdays: Climate & Environmental Change Teen Summit	2012
Presented to Abercrombie & Kent Expeditions: Ocean Acidification	2012
Presented at Mote Marine Lab REU weekly seminar on Graduate School	2010 - 2011
Volunteer at Rutgers Day: Physical Oceanography	2010 – 2013

Scientific Service

1. Frontiers in Marine Science Ocean Observation Editorial Board
2. National Offshore Wind Research and Development Consortium Wind Resource Advisory Board
3. UN Ocean Decade – Tropical Cyclone Exemplar working group
4. Underwater Glider User Group (UG2) Steering Committee
5. OceanGliders – Storms Working Group co-lead
6. Extreme Events Ocean Observations Task Team co-lead
7. New Observations and CONOPS for TCs
8. Hurricane Forecast Improvement Project (HFIP) Ocean Model Improvement Task Team (OMITT)
9. Joint WMO/IOC Technical Commission for Oceanography and Marine Meteorology: Storms Task Team and Observations Program Area
10. NOAA Joint Program Agreement on Typhoons with KIOST
11. Marine Technology Society – Poster Judging
12. Navy Task Force Ocean: Observations, Modeling, and Human Capital Working Groups
13. Reviewer: <https://publons.com/a/1361009> Frontiers in Marine Science, Journal of Physical Oceanography, Journal of Geophysical Research: Oceans, Continental Shelf Research, Estuarine, Coastal, and Shelf Science, Progress in Oceanography, Marine Geology, ICES Journal of Marine Sciences, Journal of Atmospheric and Oceanic Technology, Dynamics of Atmospheres and Oceans, and Ocean Engineering.

Rutgers University Service

1. The Chancellors Student Experience Committee – 2025 - Present
2. RU Graduate Program in Oceanography Scholastic Review Committee – 2026 - Present
3. Sloan STAR Faculty Champions – 2025 - Present
4. GH Cook Scholars Committee – 2020 – Present
5. Provost's Strategic Task Force on the Life Sciences Alliance (2023)
6. Sloan Faculty Learning Community Member 2023
7. Faculty Sponsor for Visiting Scientist Marcello Dottori (2022 – 2023)
8. Gregory Garner Faculty Review Committee (2021)
9. URGE: Unlearning Racism in Geoscience – Pod Member – 2020

10. DMCS Graduate Selections Committee 2019 – Present
11. DMCS Graduate Pre-Qualifying Exams 2019 – Present
12. DMCS Thesis Proposal Defense Chair (3 times, 2019 – Present)

Professional Societies

American Geophysical Union, American Society of Limnology and Oceanography, Marine Technology Society, The Oceanography Society, North Carolina State University Park Scholar Alumni Society, Coastal & Estuarine Research Foundation.

Graduate Students

Anthony Mendoza (MSc Thesis Advisor, 2025 – present)
Brian Buckingham (MSc Thesis Advisor, 2024 – present)
Becca Horwitz – Rutgers (PhD Thesis Advisor, 2023 - present)
Leah Hopson – Rutgers (PhD Thesis Advisor, 2022 - present)
Joseph Gradone – Rutgers (PhD Thesis Advisor, 2020 - 2024)
Sophie Scopazzi – Rutgers (MSc Thesis Advisor, 2023 – 2024)
Jake Zappala – Rutgers (MSc Thesis Advisor, Graduated 2023)
Emily Busch – Rutgers (MSc Thesis Advisor, Graduated 2023)
Malarie O’Brien – Rutgers (MSc Thesis Advisor, Graduated 2023)
Casey Jones – Rutgers (MSc Committee Member, Graduated 2022)
Samuel Coakley – Rutgers (MSc Thesis Advisor, Graduated 2022)
Sarah Murphy – Rutgers (MSc Thesis Advisor, Graduated 2021)

Graduate Committee Member

Hails Tanaka, Rutgers (PhD Committee Member, ongoing)
Samantha Alaimo, Rutgers (PhD Committee Member, ongoing)
Senam Tsei – University of Southern Mississippi (PhD Committee Member, ongoing)
Elizabeth Wright-Fairbanks – Rutgers (PhD Committee Member, Graduated 2022)
Haixing Wang – Virginia Institute of Marine Sciences (PhD Committee Member, Graduated 2021)
Julia Engdahl – Rutgers (MSc Committee Member, Graduated 2020)
Joseph Anarumo – Rutgers (MSc Committee Member, Graduated 2020)
Clifford Watkins – Rutgers (PhD Committee Member, Graduated 2020)

Graduate Student Awards

Becca Horwitz – Rutgers SuperGrad Fellowship

Leah Hopson

- MTS Oceans 2025 Student Poster Competition 2nd Place
- NCAR Bridge to Graduate Visitor Program Award
- Sloan STEM Transformations and Advancing Retention (STAR)
- GEM Fellowship
- School of Environmental and Biological Sciences (SEBS) Excellence Fellowship

Joseph Gradone

- Walter Munk Scholar and Commemorative Lecture Award
- DoD National Defense Science and Engineering Graduate Fellowship

Sam Coakley – William Lapenta – NOAA Student Internship

Sarah Murphy – Rudd Mayer Wind Power Fellow

Undergraduate Students

Oluwanifemi Kayode-Alese (NJ Wind Institute Fellow, 2024 – Present)

Lexi Leibhardt (NSF REU, 2024)

Ryan Cassin (NSF REU, 2023)
 Andrea John (NJ Wind Institute Fellow, 2023 -2024)
 Jeury Betances (Wind Institute Fellow, 2022 - 2023)
 Nikitha Shivakumar (NSF REU, 2022)
 Becca Horwitz (NSF REU, 2021)*
 Allison Porozowski (Rutgers Energy Institute Intern, 2020)
 Elisha Brumnant (NSF REU, 2019)
 Ingrid Carlson (Student Intern, 2019)
 Cassidy Gonzalez-Morabito (Challenger Glider Mission Intern, 2018)
 Liam Ramsay (Challenger Glider Mission Intern, 2018)
 Samuel Coakley (GH Cook Honors Thesis, 2016 - 2017)
 Andrenette Morrison (Challenger Glider Mission Intern, 2017)
 Benjamin Gurary (Paid Intern, 2017)
 Harrison Smith (Paid Intern, 2017)

Field Work and Research Cruises

Teledyne-Webb Slocum Glider deployments, Internationally (monthly)	2009 – present
ONR PASSENGERS Cruise 2 (remote operations 30 days), Atlantis II Seamounts	2023
ONR PASSENGERS Cruise 1 (remote operations 20 days), Atlantis II Seamounts	2022
CRATE (3 Weeks), Gulf of Lyons	2020
Jeju Island, South Korea – Typhoon Glider Deployment field campaign	2018
Amundsen Sea Polynya Cruise (ANA04B261) – R/V Araon (1 month) with the Korea Polar Research Institute (KOPRI)	2014
Long Term Ecosystem Research (LTER) Phytoplankton Component Field Team Leader, Palmer Station, Antarctica (6 months)	2011 – 2012
LTER – R/V Laurence M. Gould, Western Antarctic Peninsula (40 days)	2010 – 2011
LTER – Palmer Station, Antarctica (3 Months)	2010 – 2011
Benthic Dinoflagellate Migration – R/V Pelican, Gulf of Mexico, Florida (7 days)	2008
Student Cruise – R/V Cape Hatteras, Morehead City, NC (2 days)	2008
Upwelling and Connectivity Dynamics of EUC – M/V Sierra Negra, Galapagos, Ecuador (28 Days)	2007
Student Cruise – R/V Cape Hatteras, Morehead City, NC (2 days)	2007

Media

09/15/23 Hakai Magazine “[Scientists Eye Offshore Wind’s Effects on the Atlantic’s Crucial Cold Pool](#)”
 02/16/23 NJ.com “[In the eye of the storm](#)”
 10/8/21 The Allegheny Front “[Jersey Shore’s Fishing Industry Wonders If It Can Coexist With Industrial-Size Wind Farms](#)”
 01/30/19 – undark.org “[In Underwater Drones, A New Weapon for Hurricane Hunters](#)”
 9/20/17 –Phys.org “[New revelations from Superstorm Sandy data](#)”
 3/8/16 – Science Daily “[Why Hurricane Irene fizzled as it neared New Jersey in 2011](#)”
 7/7/14 – Science Daily: “[Antarctic climate and food web strongly linked](#)”
 10/29/13 – Popular Science: “[Underwater Gliders Gather Data To Help Predict The Next Big Storm](#)”
 12/23/13- USA Today: “[Rutgers robot’s Sandy mission may aid hurricane forecasts](#)”
 11/12/13 – Fox News Channel: “[Predicting Unpredictable Hurricanes, Superstorms](#)”
 4/25/13 – The Daily Targum: “[University students use glider to gather storm data](#)”
 11/8/12 – Newsworks: “[How Rutgers oceanographers got data from below Superstorm Sandy](#)”
 11/6/12 – Asbury Park Press: “[Rutgers researchers analyze ocean data from storm](#)”
 2/20/12 – Popular Mechanics: “[Science on ice: what its like to live and work in Antarctica](#)”

12/9/10 – Nature News Blog: “[Research trip to the Antarctic: Gliding in the Southern Ocean](#)”

Active Funded Projects

Title:

Project Role: Co-PI

Project Period: 3/1/26 – 2/28/28

Source: Office of Naval Research

Award Amount: \$

PI effort committed to project: 1 months/year

Prime applicant: Steve Anderson, Arete

Title: Robust Exploration via Multimodal Inference and Adaptive Ergodic Sampling

Project Role: Co-PI

Project Period: 9/1/25 – 8/31/28

Source: Office of Naval Research

Award Amount: \$256,542

PI effort committed to project: 1 months/year

Prime applicant: Yale University, Ian Abraham

Title: Collaborative Research: Caribbean through-flow water mass transformation processes

Project Role: Lead-PI

Project Period: 11/1/24 – 10/31/27

Source: National Science Foundation

Award Amount: \$676,526

PI effort committed to project: 1 months/year

Prime applicant: Rutgers, The State University of New Jersey

Title: Improving Loop Current Ocean Observations and Prediction

Project Role: Co-PI

Project Period: 3/1/22 – 2/28/27

Source: National Academies of Science (NAS)

Award Amount: \$690,003

PI effort committed to project: 1 months/year

Prime applicant: Texas A&M, Steve DiMarco PI

Title: MARACOOS Data to Model Comparison

Project Role: PI

Project Period: 8/1/21 – 7/31/26

Source: NOAA IOOS

Award Amount: \$660,000

PI effort committed to project: 1 months/year

Prime applicant: University of Delaware MARACOOS, PI Gerhard Kuska

Completed Projects

Title: Glider based ecological and oceanographic surveys of the New York Bight

Project Role: Co-PI

Project Period: 4/1/23 – 3/31/26

Source: New York State Energy Research and Development Authority

Award Amount: \$338,709

PI effort committed to project: 1 months/year

Prime applicant: Rutgers, The State University of New Jersey, Josh Kohut PI

Title: Unlocking Wind-Whale Co-Existence Through Artificial Intelligence

Project Role: Co-PI

Project Period: 2/1/24 – 1/31/26

Source: New Jersey Sea Grant Consortium

Award Amount: \$140,000

PI effort committed to project: 0.5 months/year

Prime applicant: Rutgers, The State University of New Jersey, Aziz Ezzat Ahmed

Title: Accelerate Improvements in Hurricane Intensity Forecasting Through Underwater Glider Field Campaigns: Mid Atlantic Regional Association Coastal Ocean Observing System (MARACOOS) Hurricane Supplemental

Project Role: PI

Project Period: 9/1/22 – 8/31/25

Source: NOAA IOOS

Award Amount: \$338,709

PI effort committed to project: 1 months/year

Prime applicant: University of Delaware MARACOOS, PI Gerhard Kuska

Title: AIRU-WRF: AI-powered Physics-based Tool for OSW Forecasting and Grid Integration

Project Role: Co-PI

Project Period: 09/1/23 – 08/31/25

Source: Department of Energy National Offshore Wind Research and Development Consortium

Award Amount: \$361,000

PI effort committed to project: 1 months/year

Prime applicant: Rutgers, The State University of New Jersey, Aziz Ezzat Ahmed

Title: New Jersey Coastal Climate Research Consortium

Project Role: Co-PI

Project Period: 06/1/2022 – 11/30/25

Source: The State of New Jersey

Award Amount: \$96,000

PI effort committed to project: 1 month/year

Prime applicant: Monmouth University

Title: Passive Acoustic Monitoring for Marine Mammal Stock Assessments using Gliders in the Caribbean: Filling a NMFS Operational Gap

Project Role: Co-PI

Project Period: 07/01/2023-06/30/2025

Source: NOAA OMAU UxS program

Award Amount: \$40,000.00

PI effort committed to project: 1 months/year

Prime applicant: University of Virgin Islands, PI Doug Wilson

Title: An investigation of potential impacts of wind turbines and foundations on the Cold Pool

Project Role: Lead PI

Project Period: 03/1/24 – 02/28/25

Source: State of New Jersey Department of Environmental Protection

Award Amount: \$97,462

PI Effort committed to project: 1 Month
Prime Applicant: Rutgers, The State University of New Jersey

Title: Glider based observations of upper ocean mixing under hurricanes

Project Role: PI

Project Period: 1/1/23 – 12/31/24

Source: NOAA OAR, CINAR

Award Amount: \$162,030

PI effort committed to project: 1 months/year

Prime applicant: Rutgers, The State University of New Jersey

Title: Targeted Stakeholder Engagement & Product Development using the RU-WRF Model Database

Project Role: Co-PI

Project Period: 7/1/23 – 6/30/24

Award Amount: \$395,483

PI effort committed to project: 1 months/year

Prime applicant: Rutgers, The State University of New Jersey, Scott Glenn

Title: Partners in Science, Operations & Maintenance of the RU-WRF Model, and Expanded Research in Support of Offshore Wind

Project Role: Co-PI

Project Period: 7/1/22 – 6/30/23

Award Amount: \$500,580

PI effort committed to project: 1 months/year

Prime applicant: Rutgers, The State University of New Jersey

Title: Assessing the impact of rapidly cycled Argo floats on operational ocean models

Project Role: PI

Project Period: 3/1/22 – 2/28/23

Source: NOAA OAR, CINAR

Award Amount: \$16,048

PI effort committed to project: 0 months/year

Prime applicant: Rutgers, The State University of New Jersey

Title: Predictions of Acoustics with Smart Experimental Networks of Gliders (PASSENGERS)

Project Role: PI

Project Period: 5/1/21 – 4/30/24

Source: Office of Naval Research

Award Amount: \$336,241

PI effort committed to project: 1 month/year

Prime applicant: Rutgers, The State University of New Jersey

Title: Onboard Processing of Slocum Glider Velocity Profiles

Project Role: PI

Project Period: 7/1/20 – 09/30/23

Source: Office of Naval Research

Award Amount: \$957,259

PI effort committed to project: 1 month/year

Prime Applicant: Rutgers, The State University of New Jersey

Title: White Paper on the Oceanographic Effects of Offshore Wind on North Atlantic Right Whale and Their Prey

Project Role: PI

Project Period: 07/01/23-04/30/24

Source: American Clean Power Association

Award Amount: \$11,348

PI effort committed to project: 1 months/year

Prime applicant: AKRF, Inc, PI Joseph Brodie

Title: Improving Forecasting of Hurricanes, Floods, and Wildfires

Project Role: Co-PI

Project Period: 7/1/20 – 6/30/23

Source: NOAA IOOS

Award Amount: \$799,437

PI effort committed to project: 1 month/year

Prime applicant: Rutgers, The State University of New Jersey, PI John Wilkin

Title: Partners in Science, Operations & Maintenance of the RU-WRF Model, and Expanded Research in Support of Offshore Wind

Project Role: Co-PI

Project Period: 7/1/22 – 6/30/23

Award Amount: \$500,580

PI effort committed to project: 1 months/year

Prime applicant: Rutgers, The State University of New Jersey

Title: RUMFS Atlantic Shores Lidar Deployment

Project Role: Co-PI

Project Period: 2/21/20 – 6/30/22

Source: EDF-Renewables, Inc.

Award Amount: \$79,697

PI effort committed to project: 0 months/year

Prime applicant: Rutgers, The State University of New Jersey

Title: Mid Atlantic Glides Supporting Hurricane Intensity Forecasts

Project Role: PI

Project Period: 7/1/20 – 6/30/22

Source: NOAA IOOS

Award Amount: \$443,749

PI effort committed to project: 1 month/year

Prime applicant: University of Delaware, PI Gerhard Kuska

Title: ARPA-E: Computationally Efficient Atmospheric-Data Driven Control Co-Design Optimization

Project Role: Collaborator

Project Period: 11/3/19 – 1/31/22

Source: US Department of Energy

Award Amount: \$1,526,872

PI effort committed to project: 0.5 months/year

Prime Applicant: Rutgers, the State University of New Jersey, PI Onur Bilgen

Title: Operations & Maintenance of the RU-WRF Model, Stakeholder Engagement and Expanded Research in Support of Offshore Wind at the request of New Jersey Board of Public Utilities

Project Role: Co-PI

Project Period: 11/1/17 – 6/30/22

Source: NJ Board of Public Utilities

Award Amount: \$1,969,189

PI effort committed to project: 1 months/year

Prime applicant: Rutgers, the State University of New Jersey, PI Scott Glenn

Title: Developing a profiling glider pH sensor for high resolution coastal ocean acidification monitoring

Project Role: Co-PI

Source: National Science Foundation OTIC

Award Amount: \$865,419

PI effort committed to project: 1 month/year

Prime applicant: Rutgers, The State University of New Jersey, PI Grace Saba

Title: Sustained Ocean Observations with underwater gliders in support of hurricane intensity forecasts

Project Role: PI

Project Period: 1/1/19 – 6/30/21

Source: NOAA OAR/US IOOS

Award Amount: \$833,537

Prime applicant: University of Delaware, PI Gerhard Kuska

Title: Coupled Atmosphere-Ocean Modeling Framework for the Mid-Atlantic Bight: WRF and ROMS

Project Role: PI

Project Period: 5/1/19 – 4/30/21

Source: NOAA IOOS

Award Amount: \$351,218

Prime applicant: Rutgers, The State University of New Jersey, PI Travis Miles

Title: Slocum Glider Integrated LISST Sensors for Particle Characterization Beneath Storms

Project Role: PI

Project Period: 7/15/16 – 7/14/17

Source: Office of Naval Research

Award Amount: \$500,000

Prime applicant: Rutgers, The State University of New Jersey, PI Travis Miles